

Lab 1: Identify pipeline components from a scenario

1. order (event) —> kafka(ingestion) —> validate order (processing) —> [orders.db](#) (storage)
2. Vendor upload(csv) —> validate schema (ingest) —> convert to parquet(processing) —> object storage(storage)
3. Clicks (events) —> logging (processing) —> log file storage

Lab 2: Classify systems as OLTP or OLAP

System	Processor	Note
Customer profile database used by customer support for updates.	OLTP	Has to be fast and current
Daily sales reporting database used by analysts for trend analysis.	OLAP	Daily reporting on historical data.
Session store for active web sessions.	OLTP	Active session means real time.
Historical analytics store for machine learning training.	OLAP	Machine learning needs a lot of historical data

Lab 3: Batch vs Streaming: classify processing

Task	Type	Reason	Tool
Generate daily sales summary for dashboard (end-of-day).	Batch	Sales summary requires multiple datapoints and is run at intervals (daily)	BI Tools, Data warehouse
Detect fraudulent transactions within 5 seconds.	Streaming	Transactions occur in real time at random frequencies. Processed immediately	Kafka
Aggregate hourly user activity for retention metrics.	Batch	Hourly aggregates.	RDBMS
Enrich clickstream events with user profile in near-real-time.	Streaming	Clicks indicate continuous dataflow.	Splink

Lab 4: ETL vs ELT decision for a use case

- Files arrive daily: ETL (ingestion, parse files, storage)
- Some fields are missing, column order varies: ELT (ingest, store, clean data)
- Analysts want to run ad-hoc queries across all products quickly. (ingest, enforce schema, store data)

Lab 5: Map sources to ingestion, storage, processing, serving

Source	Ingestion Mechanism	Storage zone	Processing
Relational orders DB (Postgres)	CDC	Bronze	Batch
Vendor CSV product files	Batch file drop	Raw	Batch
Mobile app events (high-volume)	Websockets	Raw	Stream
Product catalog in DynamoDB	API Pull	Gold	Batch

Lab 6: Design a batch e-commerce pipeline (orders)

Orchestrate [Postgres (orders) —> Loading (Parquet)—> Data Warehouse —> Analytics schema (transform) —> BI tools / Analysts (Looker, Tableau, etc.)]

Stage	Tool	Notes
Orchestration	Airflow	Daily scheduler at 2am (Cron job)
orders.db	PostgreSQL	Read data from OLTP database and convert to parquet
Loading	Parquet	Writes extracted rows to columnar files, partitioned by date
Data Warehouse	Bigquery	Loads Parquet files into a raw staging table
Analytics schema (transform)	dbt	Cleans raw data, builds fact table, aggregates daily revenue by product/region
BI	Power BI	Consumes only the analytics schema for dashboards and reporting

1. `SELECT * FROM orders WHERE updated_at >= '2026-07-12 02:00:00';`

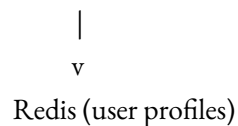
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2. INSERT INTO analytics.agg_daily_revenue_product
SELECT order_date, product_id, SUM(quantity * unit_price) AS total_revenue
FROM analytics.fct_orders
WHERE order_date = '2026-07-12'
GROUP BY order_date, product_id;

```

Lab 7: Design a streaming user activity pipeline (clickstream)

Web App -> Kafka (click-events) -> Stream Processor -> Kafka (metrics) -> Dashboard



Component	Technology	Why
Ingestion	kafka	Large volume ad velocity of clicks easily nadles
Stream processing	Flink	Low latency, stateful processing
Profile lookup	Redis	Key-value low latency lookups
Dashboard	Grafana	Reads pre-computed metrics

Lab 8: Choose ingestion tools for APIs, files, and logs

Source	Tool	Why
Third-Party REST API (hourly updates, rate-limited)	Airflow	Scheduled but low volume data ingestion. Also handles rate-limit and trail of every run
Daily FTP Drop of Large CSVs (multi-GB)	File transfer or Batch file loader.Object storage	Bulk transfer problem.
High-Volume Application Logs Across Many Servers	Kafka	High throughput, distributed, and continuous stream